

REEL

Movie Cataloging System

Team Members

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Introduction

Reel is a film catalog application that allows users to browse, search, and filter movies fetched live from the TMBD API and Supabase database. Registered users can leave ratings and write reviews, while admin users can add, edit and delete movies from the catalog, with changes persisting in the Supabase database.

1.1 System Components Tested

- App - application entry point, masthead, authentication state rendering
- AuthModal - login, registration, mode switching, Supabase authentication, role detection
- MovieCard - rendering, click handling, star ratings, admin controls, hover state
- ReviewModal - star rating selection, validation, Supabase username insert
- ReviewList - fetching, displaying and deleting reviews
- MovieCatalog - TMBD fetch, genre filtering, search, Supabase movie merge
- Modal - movie detail overlay
- AdminMovieModal - add and edit film forms
- DeleteConfirmModal - TMBD and Supabase delete logic

Testing Frameworks and Tools

- Vitest - primary test runner used across all four test files
- React Testing Library (@testing-library/react) - component rendering and DOM assertions
- vi.fn() - mock functions for Supabase client, useInView hook, and global fetch
- vi.mock() - module-level mocking to prevent network calls
- Manual testing - system-level end-to-end scenarios performed in the browser

Testing Views Applied

All three testing views are applied across this plan:

- Clear Box (CB) - tests written with full knowledge of internal logic. Used for validation rules, Supabase payload field verification, state checks, and function call assertions.
- Translucent Box (TB) - tests written with partial knowledge. Used for UI state changes, component interaction flows, and rendering.
- Opaque Box (OB) - tests written purely from the user's perspective. Used for system tests, visible rendering checks, and button/close behaviour..

2. Unit Tests

All unit tests are automated using Vitest and React Testing Library. Supabase and TMDB API are fully mocked without any use of network calls.

App Component

Tests that the application renders its core UI correctly without an authenticated user. Supabase and TMDB fetch are fully mocked.

Test ID	Method / Class	Input(s)	Expected Output	Approach	Member
UT-01-TB	App render()	No session, TMDB empty	'REEL' masthead text visible	Automated (Vitest)	Team
UT-02-TB	App render()	No session	'SIGN IN' button visible in header	Automated (Vitest)	Team
UT-03-TB	App render()	No session, TMDB empty	'A FILM CATALOG' subtitle visible	Automated (Vitest)	Team

AuthModal.jsx

Tests login/register, rendering, mode switching, Supabase auth calls, validations and the close behaviour.

Test ID	Method / Class	Input(s)	Expected Output	Approach	Member
UT-04-TB	render()	Component mount (login mode)	'Welcome back.' and email/password fields visible	Automated (Vitest)	Akash
UT-05-TB	render()	Click 'Register here'	'Join REEL.', username field, two password fields visible	Automated (Vitest)	Akash
UT-06-TB	switchMode()	Type email, switch to register	Email field cleared to empty string	Automated (Vitest)	Akash
UT-07-TB	switchMode()	Switch to register, then Sign in instead	Heading returns to 'Welcome back.'	Automated (Vitest)	Akash
UT-08-CB	handleLogin()	email + password entered	signInWithPassword called with exact credentials	Automated (Vitest)	Akash
UT-09-CB	handleLogin()	Valid credentials, role: 'user'	onSuccess called with user object and isAdmin=false	Automated (Vitest)	Akash
UT-10-CB	handleLogin()	Supabase returns error	Error '× Invalid login credentials' shown	Automated (Vitest)	Akash
UT-11-CB	handleRegister()	username: " (blank)	Error '× Username is required.' shown	Automated (Vitest)	Akash
UT-12-CB	handleRegister()	password < 6 chars	Error '× Password must be at least 6 characters.' shown	Automated (Vitest)	Akash
UT-13-CB	handleRegister()	Valid credentials submitted	supabase.auth.signUp called with email and password	Automated (Vitest)	Akash
UT-14-TB	handleRegister()	Successful sign-up	'✓ Account created! You can now log in.' shown	Automated (Vitest)	Akash
UT-15-OB	onClose()	Click [CLOSE] button	onClose callback called exactly once	Automated (Vitest)	Akash
UT-16-OB	onClose()	Click backdrop overlay	onClose callback called exactly once	Automated (Vitest)	Akash

MovieCard

Tests rendering of all visible fields, click handling, star rating logic, admin control visibility and hover/visibility state

Test ID	Method / Class	Input(s)	Expected Output	Approach	Member
UT-17-TB	render()	title: 'Inception'	Title 'Inception' visible	Automated (Vitest)	Akash
UT-18-TB	render()	year: '2010'	Year '2010' visible	Automated (Vitest)	Akash
UT-19-TB	render()	genre: ['Sci-Fi', 'Thriller']	Both genre labels visible	Automated (Vitest)	Akash
UT-20-TB	render()	rating: 8.8	'8.8' visible in score badge	Automated (Vitest)	Akash
UT-21-TB	render()	reviews: 34512	'34.5K VOTES' visible	Automated (Vitest)	Akash
UT-22-TB	render()	reviews: 450	'450 VOTES' visible	Automated (Vitest)	Akash
UT-23-TB	render()	poster URL set	img src equals poster URL	Automated (Vitest)	Akash
UT-24-TB	render()	title: 'Inception'	img alt equals 'Inception'	Automated (Vitest)	Akash
UT-25-TB	render()	index: '001'	Index stamp '001' visible	Automated (Vitest)	Akash
UT-26-TB	render()	synopsis set	Synopsis text visible on card	Automated (Vitest)	Akash
UT-27-CB	onClick()	Click card title	onClick called with full movie object	Automated (Vitest)	Akash
UT-28-CB	onClick()	Click card once	onClick called exactly once	Automated (Vitest)	Akash
UT-29-CB	star rating	rating: 8.8	5 gold spans (4 rating stars + 1 score badge)	Automated (Vitest)	Akash
UT-30-CB	star rating	rating: 10.0	6 gold spans (5 rating stars + 1 score badge)	Automated (Vitest)	Akash
UT-31-CB	star rating	rating: 2.0	2 gold spans (1 rating star + 1 score badge)	Automated (Vitest)	Akash
UT-32-CB	isAdmin prop	isAdmin: false	EDIT and DELETE buttons not in DOM	Automated (Vitest)	Akash
UT-33-TB	isAdmin prop	isAdmin: true	EDIT and DELETE buttons visible	Automated (Vitest)	Akash
UT-34-CB	onAdminEdit()	Admin clicks EDIT	onAdminEdit called; card onClick NOT called	Automated (Vitest)	Akash
UT-35-CB	onAdminDelete()	Admin clicks DELETE	onAdminDelete called with movie; onClick NOT called	Automated (Vitest)	Akash
UT-36-OB	hover state	mouseenter on card	Border contains 'rgba(46, 111, 163'	Automated (Vitest)	Akash
UT-37-OB	hover state	mouseleave after mouseEnter	Accent colour removed from border	Automated (Vitest)	Akash
UT-38-OB	useInView mock	useInView returns true	Card opacity is '1'	Automated (Vitest)	Akash
UT-39-OB	render() - favorites	currentUser logged in	Heart/favorites button visible on card	Automated (Planned)	Akash
UT-40-OB	render() - favorites	currentUser is null	Heart/favorites button not shown	Automated (Planned)	Akash
UT-41-CB	handleAddFavorite()	Click unfavorited heart	Supabase insert called with user_id and movie_id	Automated (Planned)	Akash
UT-42-CB	handleRemoveFavorite()	Click favorited heart	Supabase delete called with correct user_id	Automated (Planned)	Akash
UT-43-TB	handleAddFavorite()	Successful insert	Heart toggles to filled/favorited state	Automated (Planned)	Akash
UT-44-TB	handleRemoveFavorite()	Successful delete	Heart toggles back to unfilled state	Automated (Planned)	Akash

ReviewModal

Tests rendering, star interaction, all three validation rules, the Supabase insert payloads, loading state, error display, and close behaviour.

Test ID	Method / Class	Input(s)	Expected Output	Approach	Member
UT-45-TB	render()	Movie title 'War Machine'	'War Machine' in heading	Automated (Vitest)	Akash
UT-46-TB	render()	Component mount	5 star buttons rendered	Automated (Vitest)	Akash
UT-47-TB	render()	Component mount	Textarea placeholder visible	Automated (Vitest)	Akash
UT-48-TB	render()	Component mount	SUBMIT REVIEW button rendered	Automated (Vitest)	Akash
UT-49-TB	render()	Component mount	Close button rendered	Automated (Vitest)	Akash
UT-50-CB	render()	Type 'Great film!' in textarea	'11 chars' count shown	Automated (Vitest)	Akash
UT-51-CB	setRating()	Component mount	All 5 stars have aria-pressed='false'	Automated (Vitest)	Akash
UT-52-CB	setRating()	Click 3rd star	3rd star has aria-pressed='true'	Automated (Vitest)	Akash
UT-53-CB	setRating()	Click 2nd then 5th star	5th: 'true'; 2nd: 'false'	Automated (Vitest)	Akash
UT-54-CB	handleSubmit()	rating=0, valid text	Error '× Please select a star rating.' shown	Automated (Vitest)	Akash
UT-55-CB	handleSubmit()	rating=5, text=""	Error '× Review cannot be empty.' shown	Automated (Vitest)	Akash
UT-56-CB	handleSubmit()	rating=3, text='Too short'	Error '× Review must be at least 10 characters.' shown	Automated (Vitest)	Akash
UT-57-CB	handleSubmit()	Validation fails	Supabase insert NOT called	Automated (Vitest)	Akash
UT-58-CB	handleSubmit()	rating=3, valid text	Insert called with user_id, movie_id, title, rating, review_text, username	Automated (Vitest)	Akash
UT-59-CB	handleSubmit()	currentUser.username undefined	username falls back to currentUser.email	Automated (Vitest)	Akash
UT-60-CB	handleSubmit()	movie.id = 42 (number)	movie_id stored as string '42'	Automated (Vitest)	Akash
UT-61-CB	handleSubmit()	Successful insert	onSave() called once	Automated (Vitest)	Akash
UT-62-TB	handleSubmit()	Insert in flight	Button shows 'SUBMITTING...'	Automated (Vitest)	Akash
UT-63-CB	handleSubmit()	Supabase returns error	'× DB connection failed' shown	Automated (Vitest)	Akash
UT-64-CB	handleSubmit()	Text with leading/trailing spaces	review_text trimmed before insert	Automated (Vitest)	Akash
UT-65-OB	onClose()	Click CANCEL	onClose called once	Automated (Vitest)	Akash
UT-66-OB	onClose()	Click [CLOSE]	onClose called once	Automated (Vitest)	Akash
UT-67-OB	onClose()	Click modal inner content	onClose NOT called	Automated (Vitest)	Akash

ReviewsList

Tests the review list fetch, display, username rendering, empty state, review count header, button visibility, and delete behaviour.

Test ID	Method / Class	Input(s)	Expected Output	Approach	Member
UT-68-TB	fetchReviews()	Supabase not yet resolved	'LOADING REVIEWS...' visible	Automated (Vitest)	Akash
UT-69-TB	fetchReviews()	Two reviews returned	Both review_text values visible	Automated (Vitest)	Akash
UT-70-CB	fetchReviews()	Reviews with username field set	Username 'akash' and 'elias' visible	Automated (Vitest)	Akash
UT-71-CB	fetchReviews()	username: null in review	'Anonymous' displayed	Automated (Vitest)	Akash
UT-72-OB	fetchReviews()	Empty array returned	'No reviews yet. Be the first...' visible	Automated (Vitest)	Akash
UT-73-TB	fetchReviews()	2 reviews returned	Header shows 'REVIEWS (2)'	Automated (Vitest)	Akash
UT-74-OB	render()	currentUser logged in	'+ WRITE A REVIEW' button visible	Automated (Vitest)	Akash
UT-75-OB	render()	currentUser is null	'+ WRITE A REVIEW' button not in DOM	Automated (Vitest)	Akash
UT-76-TB	render()	Click Write a Review	ReviewModal textarea appears	Automated (Vitest)	Akash
UT-77-CB	handleDelete()	2 reviews; user owns r1 only	Exactly 1 DELETE button visible	Automated (Vitest)	Akash
UT-78-CB	handleDelete()	User logged in as owner of r2	Exactly 1 DELETE button visible for r2	Automated (Vitest)	Akash
UT-79-CB	handleDelete()	Click DELETE on own review	Supabase delete called with ('id', 'r1')	Automated (Vitest)	Akash
UT-80-TB	handleDelete()	Successful delete	Deleted review removed from DOM	Automated (Vitest)	Akash
UT-81-TB	handleDelete()	Delete one of two reviews	Other review still visible	Automated (Vitest)	Akash

AdminMovieModal

Unit tests for the add/edit film form are currently covered at integration and system level. Dedicated unit tests are planned for the next iteration.

Test ID	Method / Class	Input(s)	Expected Output	Approach	Member
UT-82-TB	render() - add mode	No editMovie prop	'Add New Film' heading; all fields empty	Automated (Planned)	Abijit
UT-83-TB	render() - edit mode	editMovie with title 'Inception'	'Editing: Inception' heading; fields pre-filled	Automated (Planned)	Abijit
UT-84-CB	handleSave()	title: " (blank)	Error 'X Title is required.' shown	Automated (Planned)	Abijit
UT-85-CB	handleSave()	year: 'notayear'	Error 'X Valid year is required.' shown	Automated (Planned)	Abijit
UT-86-CB	handleSave() - insert	Valid fields; genre 'Action, Drama'	Insert called with genre: ['Action', 'Drama']	Automated (Planned)	Abijit
UT-87-CB	handleSave() - update	editMovie.supabaseId = 'abc-123'	Update called with id 'abc-123'; insert NOT called	Automated (Planned)	Abijit
UT-88-CB	handleSave()	Title with extra spaces	title trimmed before saving	Automated (Planned)	Abijit
UT-89-TB	handleSave()	Successful save	onSave() called once; modal closes	Automated (Planned)	Abijit

UT-90-TB	handleSave()	Request in flight	Button shows 'SAVING...'	Automated (Planned)	Abijit
UT-91-OB	onClose()	Click CANCEL	onClose called once	Automated (Planned)	Abijit

AdminMovieModal

The DeleteConfirmModal handles two deletion paths and needs dedicated unit coverage. Tests are planned for the next iteration.

Test ID	Method / Class	Input(s)	Expected Output	Approach	Member
UT-92-TB	render()	movie.title = 'Inception'	'Remove "Inception"?' heading shown	Automated (Planned)	Abijit
UT-93-CB	handleDelete() - Supabase	fromSupabase=true, supabaseId='abc'	Supabase delete called with id 'abc'	Automated (Planned)	Abijit
UT-94-CB	handleDelete() - TMDB	fromSupabase=false, movie.id=42	Insert called with tmdb_id: 42, deleted: true	Automated (Planned)	Abijit
UT-95-CB	handleDelete()	Successful delete	onConfirm called with deleted movie object	Automated (Planned)	Abijit
UT-96-TB	handleDelete()	Request in flight	Button shows 'DELETING...'	Automated (Planned)	Abijit
UT-97-CB	handleDelete()	Supabase returns error	Error message shown; onConfirm NOT called	Automated (Planned)	Abijit
UT-98-OB	onClose()	Click CANCEL	onClose called once	Automated (Planned)	Abijit

FavouritesPage

The Favorites feature is scheduled for the next sprint. TDD red tests have already been written. The following unit tests will be implemented alongside the feature build.

Test ID	Method / Class	Input(s)	Expected Output	Approach	Member
UT-99-OB	FavoritesPage render()	User has 2 favorited films	Both movie cards visible on favorites page	Automated (Planned)	Akash
UT-100-OB	FavoritesPage render()	User has no favorites	'No favorites yet' message shown	Automated (Planned)	Akash
UT-101-CB	FavoritesPage fetch()	Component mounts	Supabase select called with correct user_id	Automated (Planned)	Akash

3. Integration Tests

Integration tests evaluate interactions between two or more components or services. These verify that data flows correctly between the frontend and Supabase, and that components communicate as expected.

Test ID	Components	Preconditions	Steps	Expected Result	Member
IT-01-CB	AuthModal + Supabase Auth + profiles	Mocks set up; AuthModal rendered	1. Enter email/password 2. Click SIGN IN 3. Check signInWithPassword call 4. Check profiles fetch	signInWithPassword called; profile fetched; onSuccess fires with correct user and isAdmin flag	Akash

IT-02-CB	AuthModal + profiles table (register)	Sign-up mock returns new user id	1. Switch to register 2. Fill fields 3. Submit 4. Check profiles insert	profiles.insert called with id, username, role='user'	Akash
IT-03-TB	ReviewModal + ReviewsList + Supabase	User logged in; ReviewsList rendered	1. Click + WRITE A REVIEW 2. Select stars and enter text 3. Submit 4. Check list refresh	Supabase insert called; ReviewsList re-fetches; new review appears in list	Akash
IT-04-TB	ReviewsList + Supabase delete	Two reviews in DB; user owns one	1. Render ReviewsList 2. Click DELETE on own review 3. Check delete call 4. Check list	delete called with correct id; review removed; other review remains	Akash
IT-05-CB	MovieCatalog + TMDB API (genre filter)	TMDB mock returns movies	1. Render MovieCatalog 2. Observe /movie/popular fetch 3. Click a genre 4. Observe /discover/movie fetch	Popular movies on load; genre click triggers TMDB discover endpoint fetch	Cameron
IT-06-TB	MovieCatalog + Supabase movies table	Supabase has one admin-added film	1. Render MovieCatalog 2. Check Supabase fetch 3. Check grid order	Supabase film shown first; TMDB films follow; no duplicates	Akash
IT-07-TB	AdminMovieModal + Supabase movies	Admin logged in; Add Film modal open	1. Fill all fields 2. Click ADD FILM 3. Check Supabase insert	Insert called with correct formData; modal closes; new film in grid	Abijit
IT-08-OB	Modal + ReviewsList	User logged in; movie card clicked	1. Click movie card 2. Modal opens 3. Check reviews section	ReviewsList renders inside Modal; reviews fetched; Write a Review button visible	Akash
IT-09-TB	MovieCatalog + Search + TMDB	MovieCatalog rendered	1. Type 'Inception' in search 2. Submit 3. Check fetch URL	Fetch to /search/movie?query=Inception; page resets to 1; genre resets to ALL	Abijit
IT-10-OB	DeleteConfirmModal + Supabase + MovieCatalog	Admin logged in; TMDB movie visible	1. Click DELETE on TMDB movie 2. Confirm 3. Check insert with deleted:true 4. Check grid	TMDB card removed; deleted marker inserted to Supabase; hidden on reload	Abijit
IT-11-CB	MovieCard + Supabase favorites (add)	User logged in; film not yet favorited	1. Render MovieCard 2. Click heart button 3. Check Supabase insert 4. Check heart state	Insert called with user_id and movie_id; heart toggles to filled state	Akash
IT-12-CB	FavoritesPage + Supabase favorites	User logged in; 2 films favorited in DB	1. Render FavoritesPage 2. Check Supabase select 3. Verify both films render	select called with user_id; both favorited film cards displayed	Akash
IT-13-TB	ReviewsList + ReviewModal (count refresh)	Film has 1 existing review	1. Open modal 2. Submit new review 3. Check ReviewsList refreshes	ReviewsList re-fetches; count updates from (1) to (2); new review visible	Akash
IT-14-CB	DeleteConfirmModal + MovieCatalog (Supabase film)	Admin logged in; Supabase film in grid	1. Click DELETE on Supabase card 2. Confirm 3. Check Supabase delete 4. Check grid	Hard delete called; film removed from supabaseMovies state; card disappears	Akash
IT-15-TB	AdminMovieModal + Modal (edit flow)	Admin logged in; movie modal open	1. Click EDIT MOVIE in modal 2. Modal closes 3. AdminMovieModal opens pre-filled 4. Save	Fields pre-filled with current data; save updates Supabase; card reflects changes	Abijit

4. System Tests

System tests validate complete end-to-end user workflows from the user's perspective. All system tests are performed manually in the browser against the live application connected to Supabase and the TMDB API.

Test ID	Scenario	Preconditions	Steps	Expected Outcome	Member
ST-01-OB	Register and log in	App running; user not registered	1. Click SIGN IN 2. Click Register here 3. Enter details 4. Submit 5. Log in	Account created in Supabase; login succeeds; username in header	Akash
ST-02-OB	Browse popular films	App loaded	1. Open app 2. Scroll grid 3. Click a movie card	Popular movies load from TMDB; modal opens with full details and reviews	Team
ST-03-OB	Search for a film	App loaded	1. Type 'The Dark Knight' 2. Click SEARCH 3. Check results	TMDB search results shown; CLEAR button appears; CLEAR returns to popular	Team
ST-04-OB	Filter by genre	App loaded	1. Click ACTION genre 2. Browse 3. Switch genre 4. Click ALL	TMDB discover fetches Action movies; pagination works; ALL restores popular	Team
ST-05-OB	Write a review	Logged in; movie modal open	1. Click + WRITE A REVIEW 2. Select 4 stars 3. Enter 10+ chars 4. Submit	Review saved to Supabase with username; appears in list immediately	Akash
ST-06-OB	Delete own review	Logged in; has at least one review	1. Open film modal 2. Locate own review 3. Click DELETE	Review removed from Supabase and DOM; other users' reviews unaffected	Akash
ST-07-TB	Admin adds a film	Admin logged in	1. Click + ADD FILM 2. Fill all fields 3. Click ADD FILM	Film saved to Supabase; appears at top of grid with EDIT/DELETE controls	Team
ST-08-TB	Admin edits a TMDB film	Admin logged in	1. Click EDIT on TMDB card 2. Modify synopsis 3. Save	Supabase row inserted with tmdb_id; TMDB card hidden; edited version shown	Team
ST-09-TB	Admin deletes a TMDB film	Admin logged in	1. Click DELETE on TMDB card 2. Confirm	Deleted marker in Supabase; film hidden; stays hidden on reload	Team
ST-10-OB	Session persists on reload	User logged in	1. Log in 2. Refresh browser	Session restored from Supabase; username shown; no re-login required	Akash
ST-11-OB	Save and view favorites	User logged in; browsing catalog	1. Click heart on a film 2. Navigate to My Favorites 3. Verify film appears	Film saved to Supabase; appears on favorites page; heart shown as filled	Akash
ST-12-OB	Remove from favorites	User has at least one favorited film	1. Open favorites page 2. Click filled heart 3. Verify removal	Film removed from Supabase; disappears from favorites list immediately	Akash
ST-13-OB	Unauthenticated users cannot write reviews	App loaded; no user logged in	1. Open a movie modal 2. Check reviews section	No Write a Review button shown; no way to submit a review without logging in	Akash
ST-14-OB	User cannot delete another user's review	Two users have reviewed same film	1. Log in as user A 2. Open film modal 3. Check DELETE button visibility	DELETE absent on user B's review; present only on user A's own review	Akash
ST-15-TB	Pagination within a genre filter	App loaded; genre active	1. Click ACTION genre 2. Navigate to page 2 3. Navigate back to page 1	Each page fetches correct Action films from TMDB; no cross-genre bleed	Team

5. Demonstration of Coverage

Suite / File	CB	TB	OB	Total
App.test (App component)	0	3	0	3
authModal.test (AuthModal)	8	3	2	13
movieCard.test (MovieCard)	10	8	6	24
Reviews.test (ReviewModal + List)	14	14	10	38
AdminMovieModal (planned)	6	3	1	10
DeleteConfirmModal (planned)	4	2	1	7
Favorites feature (planned)	4	3	2	9
Integration Tests	3	4	3	10
System Tests	0	3	7	10
TOTAL	49	43	32	134

Mapping to User Stories

User Story	Test IDs	Level	Views
As a user, I want to log in	UT-04 to UT-10, UT-15, UT-16, IT-01, ST-01, ST-10	Unit, Integration, System	CB, TB, OB
As a new user, I want to register	UT-05, UT-11 to UT-14, IT-02, ST-01	Unit, Integration, System	CB, TB, OB
As a user, I want to browse and search films	UT-01 to UT-03, UT-17 to UT-28, IT-05, IT-09, ST-02 to ST-04	Unit, Integration, System	CB, TB, OB
As a registered user, I want to write reviews	UT-45 to UT-81, IT-03, IT-04, IT-08, IT-13, ST-05, ST-06, ST-13, ST-14	Unit, Integration, System	CB, TB, OB
As an admin, I want to add, edit, delete films	UT-82 to UT-98, IT-06, IT-07, IT-10, IT-14, IT-15, ST-07 to ST-09	Unit, Integration, System	CB, TB, OB
As a user, I want to save favorites (planned)	UT-39 to UT-44, UT-99 to UT-101, IT-11, IT-12, ST-11, ST-12	Unit, Integration, System	CB, TB, OB

High Risk Components & Coverage Rationale

- AuthModal (13 unit + 2 integration) - handles all authentication and role assignment. Failures here lock all users out. CB tests verify exact Supabase payloads, OB tests verify what users observe.
- ReviewModal + ReviewsList (37 unit + 3 integration planned) - directly reads and writes Supabase. Username fallback, validation rules, and delete only own review logic are critical for data integrity.
- MovieCatalog TMDB + Supabase merge (4 integration + 3 system planned) - the most complex component. Merging two data sources, filtering deleted/overridden films, and the genre refetch system require integration level verification.
- Favorites feature (9 unit + 2 integration + 2 system - all planned) - new feature with TDD red tests already written. Full coverage planned before the green phase begins.

Testing Views Applied

- Clear Box (CB) - 49 tests (35 implemented, 14 planned). Verifies Supabase payload fields, validation boundaries, state mutations, and function call assertions.
- Translucent Box (TB) - 43 tests (35 implemented, 8 planned). Verifies UI state changes, button labels, mode switching, and rendering updates.
- Opaque Box (OB) - 32 tests (28 implemented, 4 planned). Verifies observable user facing behaviour such as visible text, hover effects, close button logic.

Known Gaps and Limitations

The following known gaps have been identified in the current plan, along with how each will be resolved in the next iteration.

Planned but Not Yet Implemented: Approximately 30 of 134 tests (around 22%) are marked "Planned" and not yet written, covering AdminMovieModal (UT-82 to UT-91), DeleteConfirmModal (UT-92 to UT-98), and FavoritesPage (UT-99 to UT-101, IT-11, IT-12, ST-11, ST-12). All will be implemented in the next iteration alongside the FavoritesPage green phase.

Negative Integration Testing: Integration tests focus on success flows and controlled mock failures. There are no tests for infrastructure level failures such as Supabase possibly becoming unreachable during a session or TMDB returning a malformed response. Error path test types will be added to IT-03, IT-05, and IT-07 in the next iteration.

Unassigned Test Ownership: "Team" currently listed as the assigned. Individual assignments will be finalized before the testing phase begins.

Performance and Cross Browser Testing: No performance testing exists for the TMDB/Supabase merge in MovieCatalog, and system tests were conducted in a single desktop browser environment. Both are accepted gaps for this iteration given project scope.